Designing a CRISPR-Cas9 Guide RNA for BCL11A: A Simple Walkthrough

Prepared for the CRISPR-Cas9 review paper. Tools accessed on 24 Oct 2026.

1. What we wanted to find out

Our review paper makes one big claim about CRISPR-Cas9: it is **simple and efficient to design**. Older tools such as ZFNs and TALENs need a brand-new protein to be engineered for every DNA target. CRISPR-Cas9 does not. You only need to choose a short stretch of about 20 letters (a **spacer**, the main part of the **guide RNA or gRNA**) that matches the gene you want to cut.

So our objective was to **prove this claim with a real example**, using only free web tools and a laptop. We picked a human gene, designed a guide for it, checked how well it should work, and checked whether it might cut the wrong places.

2. A quick plain-language background

- **Cas9** is the molecular scissors. **SpCas9** is the most common version, taken from *Streptococcus pyogenes*.
- The **guide RNA** is the GPS. Its ~20 letter spacer tells Cas9 where to cut.
- Cas9 will only cut if the DNA next to the target has a short tag called a **PAM**. For SpCas9 the PAM is **NGG** (any letter, then two Gs).
- Cas9 cuts about **3 letters before the PAM**. The cell then repairs the break, often with small mistakes (insertions or deletions) that can switch the gene off. This is called a **knock-out**.
- **Off-targets** are other places in the genome that look similar to the guide and might get cut by accident. We want as few as possible. A site that differs from the guide by **0-2 letters (mismatches)** is the most worrying, so we look for none of those.

3. Our choices

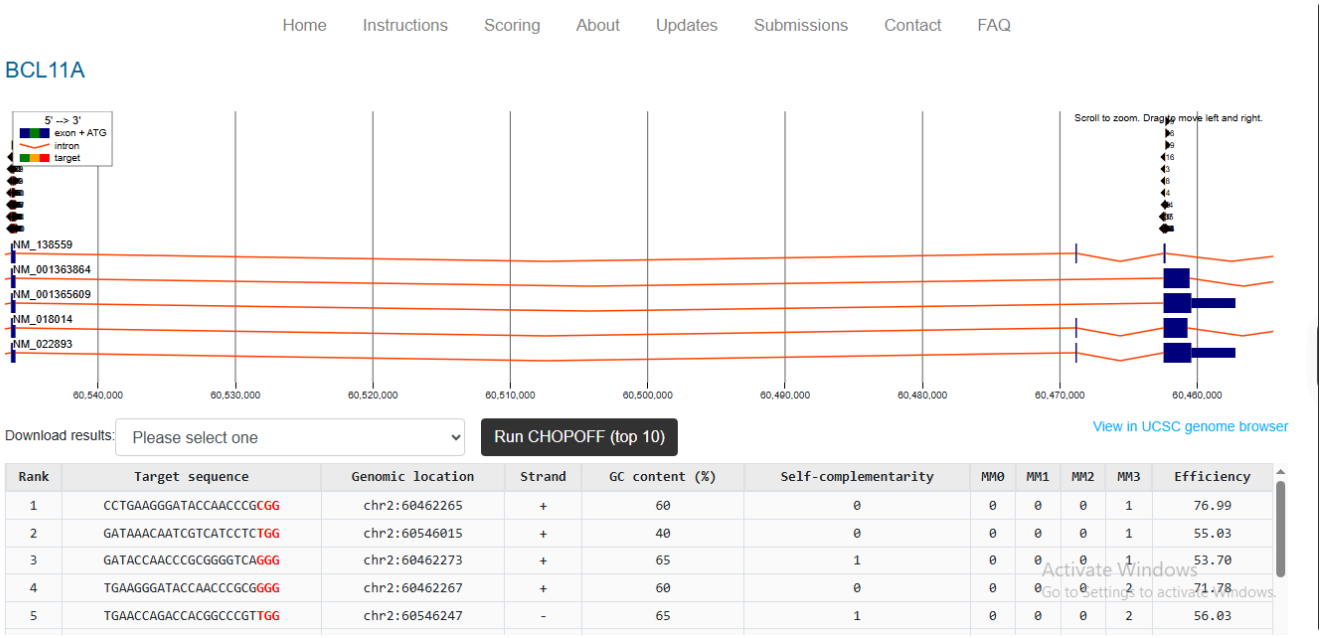
Item	What we chose
Gene	BCL11A (human)
Genome	Human, version hg38
Nuclease	SpCas9 (PAM = NGG)
Goal	Knock-out
Tools	CHOPCHOP and CRISPOR (both free, run in the browser, nothing to download)

We chose BCL11A because it is well known in genome editing. An important note: the approved therapy Casgevy, as far as we know, edits an *enhancer* of BCL11A, not the gene's coding exon. Our guide targets a coding exon, so this is an **illustrative knock-out design**, not the therapeutic guide.

4. Step by step: what we did and saw

Step A: CHOPCHOP finds possible guides (Figure 1)

We typed "BCL11A", chose Homo sapiens (hg38), CRISPR/Cas9 and knock-out, and clicked Find Target Sites. CHOPCHOP drew the gene map and listed ranked guides.



Rank	Target sequence	Genomic location	Strand	GC content (%)	Self-complementarity	MM0	MM1	MM2	MM3	Efficiency
1	CCTGAAGGGATACCAACCCGCGG	chr2:60462265	+	60	0	0	0	0	1	76.99
2	GATAAACAATCGTCATCCTCTGG	chr2:60546015	+	40	0	0	0	0	1	55.03
3	GATACCAACCCGCGGGGTCAGGG	chr2:60462273	+	65	1	0	0	0	1	53.70
4	TGAAGGGATACCAACCCGCGGGG	chr2:60462267	+	60	0	0	0	0	2	71.78
5	TGAACCAGACCACGGCCCGTTGG	chr2:60546247	-	65	1	0	0	0	2	56.03

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Figure 1. CHOPCHOP results for BCL11A (hg38, CRISPR/Cas9, knock-out).

What you are looking at: The top half is a map of the BCL11A gene. The orange lines are introns, the dark blue blocks are exons, and the small black arrows are the places where possible guides fit. The table underneath lists those guides from best to worst. Each row gives the 20-letter target sequence with its PAM in red, where it sits on chromosome 2, its GC content, how likely it is to fold on itself, how many look-alike sites exist in the genome (MM0 to MM3), and a predicted efficiency score. Rank 1 is the best: efficiency 76.99, GC 60%, and no look-alike sites with 0, 1 or 2 mismatches.

The top-ranked guide (Rank 1) was:

- Spacer: `CCTGAAGGGATACCAACCCG`
- PAM: `CGG`
- Location: chr2:60462265, forward strand
- GC content: 60% (a healthy range is roughly 40-70%)
- Self-complementarity: 0 (it will not fold back on itself)
- Predicted efficiency: 76.99
- Off-targets with 0, 1 or 2 mismatches: none. One site with 3 mismatches.

Step B: Looking closely at Rank 1 (Figure 2)

Clicking Rank 1 opened a zoomed view. It shows the guide sitting inside a coding exon that is shared by most BCL11A transcripts. The blue mark on the guide is roughly where Cas9 would cut.

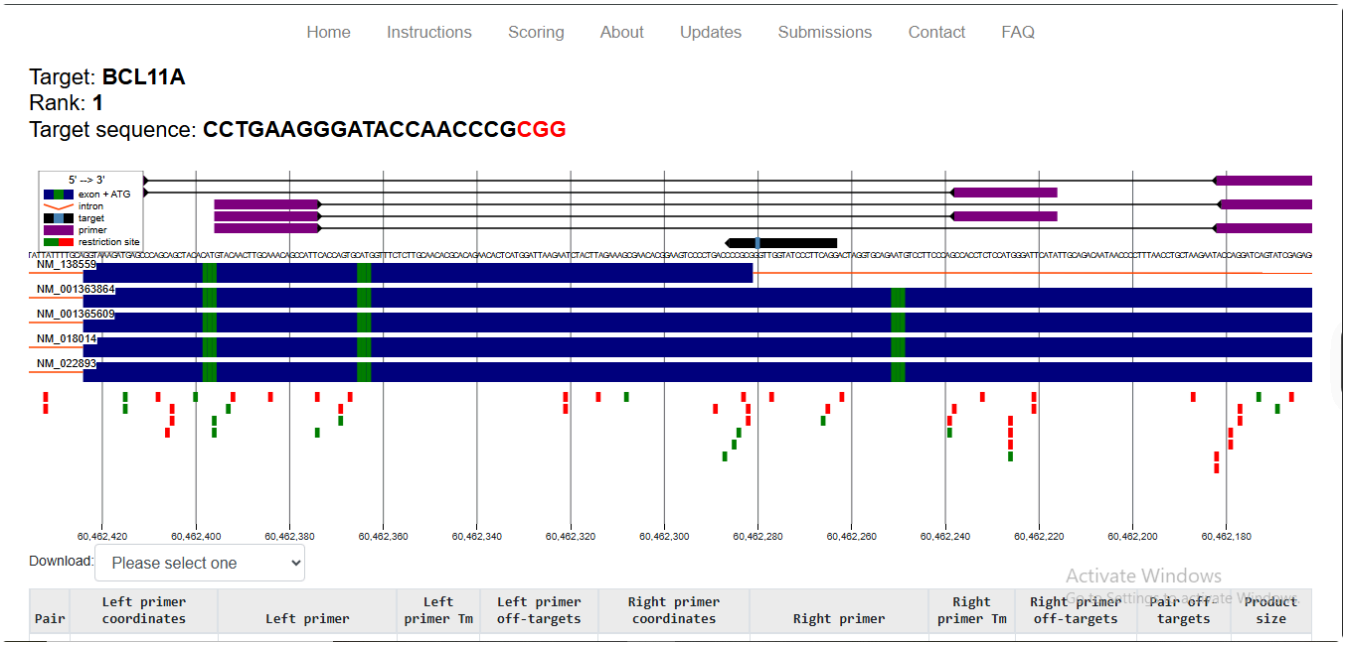


Figure 2. The Rank 1 guide `CCTGAAGGGATACCAACCCG` (PAM CGG) shown inside the gene.

What you are looking at: This is a close-up of the guide. The black bar with the blue mark is our guide, and the blue mark is roughly where Cas9 would cut, about 3 letters before the PAM. The dark blue bands are the exons of the different BCL11A versions (the NM_ numbers), so the guide sits inside a coding exon. The purple bars are suggested PCR primers around the cut, and the green and red marks are restriction enzyme sites. In simple words, this picture shows that our guide lands in the part of the gene that codes for protein, which is what we want for a knock-out.

Step C: Primers and the predicted repair result (Figure 3)

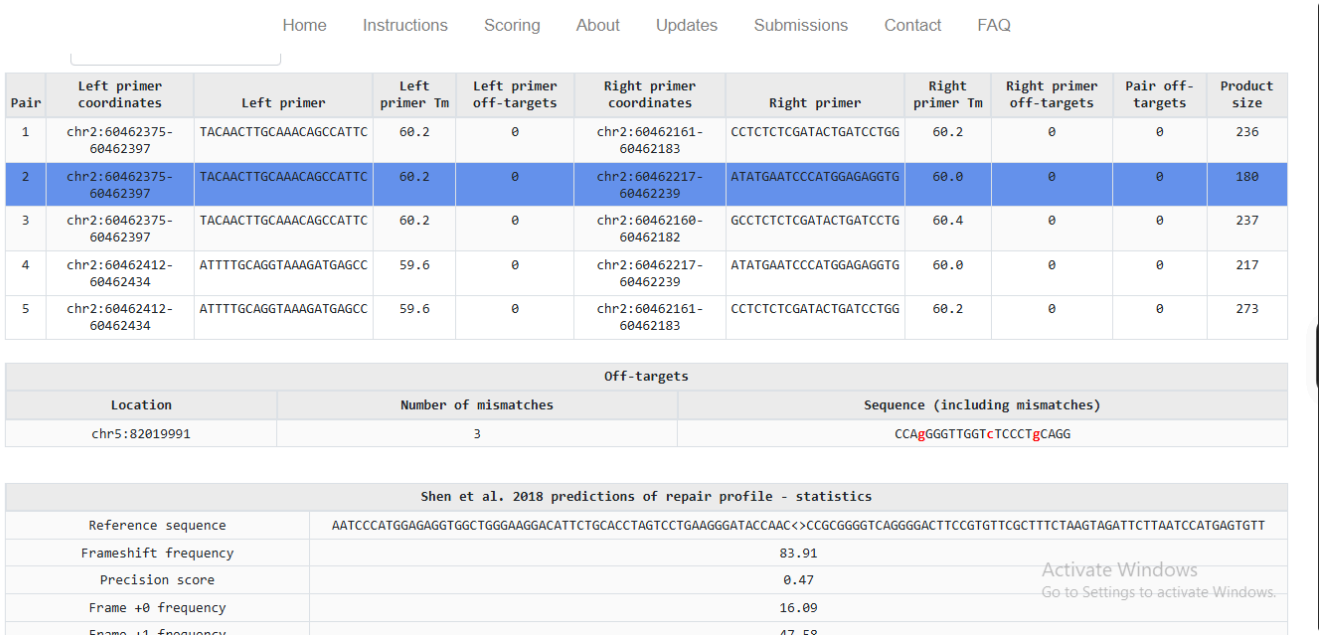
CHOPCHOP also suggested PCR primers to check the edit in a lab. We chose **Pair 1** because both primers have the same melting temperature (60.2 °C), neither has predicted off-targets, and the product (236 bp) is a good size for a simple test such as a T7E1 assay.

- Left primer: `TACAACTTGCAAACAGCCATTC`
- Right primer: `CCTCTCTCGATACTGATCCTGG`

A computer model (Shen et al., 2018) then predicted how the cell would repair the cut:

Predicted outcome	Value
Frameshift outcomes	83.91%
In-frame outcomes (+0)	16.09%
+1 frame / +2 frame	47.58% / 36.33%
1-bp insertion	6.46%
Most likely single outcome	19.10% (a deletion)
Deletions using microhomology	77.42%
Precision score	0.47

In simple words: **most predicted repairs would scramble the gene's reading frame**, which is exactly what you want for a knock-out. Roughly 16% of outcomes keep the frame and might leave a working protein, so real experiments would need checking by sequencing.



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Pair	Left primer coordinates	Left primer	Left primer Tm	Left primer off-targets	Right primer coordinates	Right primer	Right primer Tm	Right primer off-targets	Pair off-targets	Product size
1	chr2:60462375-60462397	TACAACTTGCAAACAGCCATTC	60.2	0	chr2:60462161-60462183	CCTCTCTCGATACTGATCCTGG	60.2	0	0	236
2	chr2:60462375-60462397	TACAACTTGCAAACAGCCATTC	60.2	0	chr2:60462217-60462239	ATATGAATCCCATGGAGAGGTG	60.0	0	0	180
3	chr2:60462375-60462397	TACAACTTGCAAACAGCCATTC	60.2	0	chr2:60462160-60462182	GCCTCTCTCGATACTGATCCTG	60.4	0	0	237
4	chr2:60462412-60462434	ATTTTGCAGGTAAAGATGAGCC	59.6	0	chr2:60462217-60462239	ATATGAATCCCATGGAGAGGTG	60.0	0	0	217
5	chr2:60462412-60462434	ATTTTGCAGGTAAAGATGAGCC	59.6	0	chr2:60462161-60462183	CCTCTCTCGATACTGATCCTGG	60.2	0	0	273

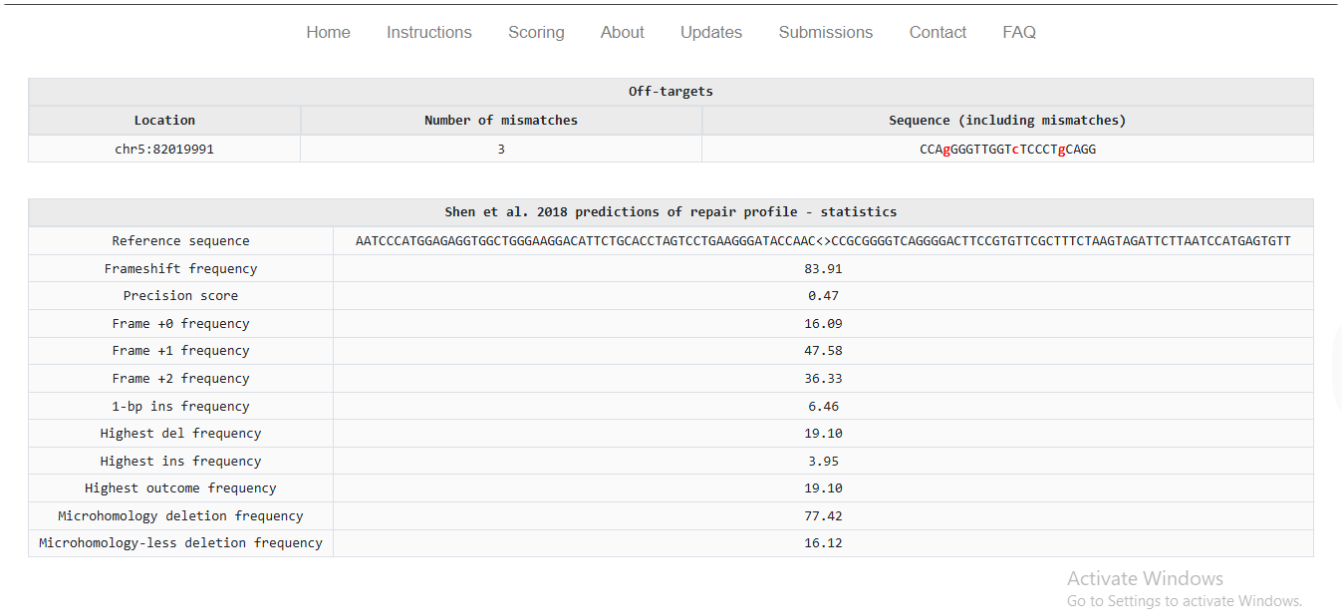
Off-targets		
Location	Number of mismatches	Sequence (including mismatches)
chr5:82019991	3	CCAgGGGTTGGTcTCCCTgCAGG

Shen et al. 2018 predictions of repair profile - statistics	
Reference sequence	AATCCCATGGAGAGGTGGCTGGGAAGGACATTCTGCACCTAGTCCTGAAGGGATACCAAC<>CCGCGGGGTCAGGGGACTTCCGTGTTCGCTTTCTAAGTAGATTCTTAATCCATGAGTGTT
Frameshift frequency	83.91
Precision score	0.47
Frame +0 frequency	16.09
Frame +1 frequency	47.58

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Figure 3A. Primer pairs suggested by CHOPCHOP and the single predicted off-target site.

What you are looking at: The table lists five primer pairs that could be used to copy the region around the cut and check whether it was edited. We chose Pair 1: both primers melt at 60.2 °C, neither has off-targets, and the product is 236 bp long. Under the primer table, the Off-targets box shows the only look-alike site CHOPCHOP found, on chromosome 5, and it differs from our guide in 3 letters, so it is unlikely to be cut.



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Off-targets		
Location	Number of mismatches	Sequence (including mismatches)
chr5:82019991	3	CCAgGGGTTGGTcTCCCTgCAGG

Shen et al. 2018 predictions of repair profile - statistics	
Reference sequence	AATCCCATGGAGAGGTGGCTGGGAAGGACATTCTGCACCTAGTCCTGAAGGGATACCAAC<>CCGCGGGGTCAGGGGACTTCCGTGTTCGCTTTCTAAGTAGATTCTTAATCCATGAGTGTT
Frameshift frequency	83.91
Precision score	0.47
Frame +0 frequency	16.09
Frame +1 frequency	47.58
Frame +2 frequency	36.33
1-bp ins frequency	6.46
Highest del frequency	19.10
Highest ins frequency	3.95
Highest outcome frequency	19.10
Microhomology deletion frequency	77.42
Microhomology-less deletion frequency	16.12

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Figure 3B. Predicted repair outcomes after Cas9 cuts at the Rank 1 site.

What you are looking at: This table comes from a computer model that guesses how the cell will repair the cut. The key number is the frameshift frequency, 83.91%: most predicted repairs add or remove letters in a way that scrambles the reading frame and switches the gene off. About 16.09% of outcomes keep the frame and might leave a working protein. The model also predicts that 77.42% of deletions use short matching stretches of DNA (microhomology), which links back to the repair-pathways part of the review. These are predictions, not lab results.

Step D: A second opinion from CRISPOR (Figure 4)

To avoid trusting a single tool, we pasted the same exon sequence into CRISPOR (hg38, 20bp-NGG SpCas9). CRISPOR found 18 possible guides in our 120 bp input. Its coordinates matched CHOPCHOP exactly (chr2:60462265), so both tools were looking at the same spot.

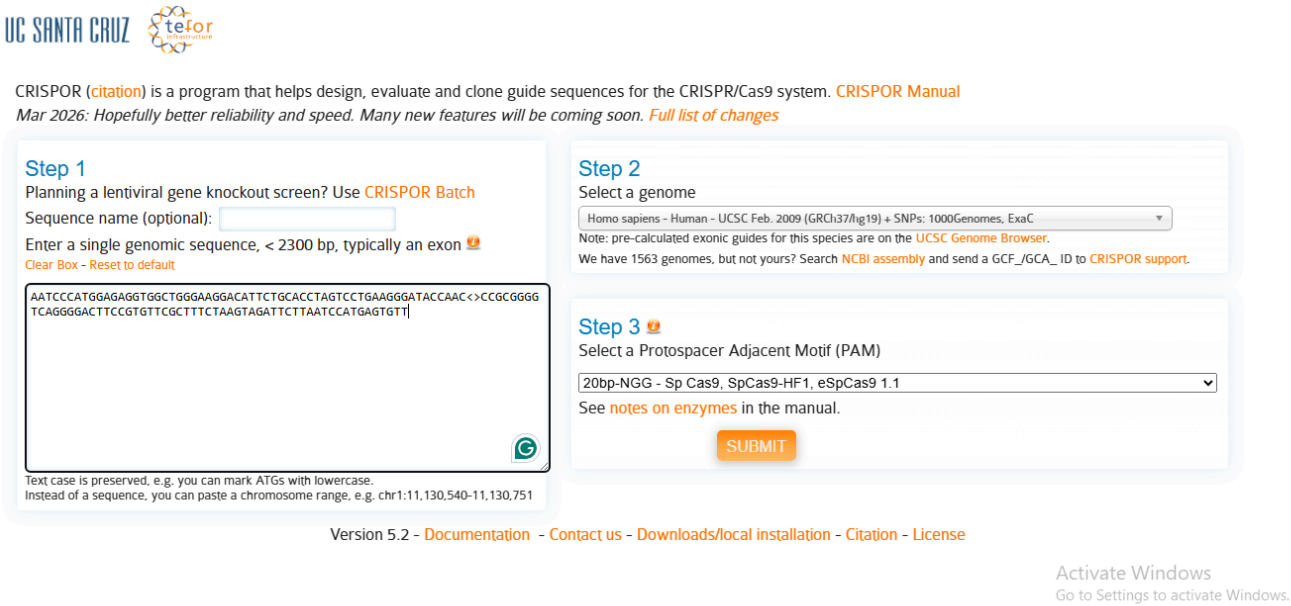


Figure 4A. The CRISPOR input page with our exon sequence pasted in.

What you are looking at: Step 1 holds the DNA sequence copied from CHOPCHOP, Step 2 picks the genome, and Step 3 picks the PAM type (NGG for SpCas9). Note for the paper: this screenshot was taken before two fixes. The genome box still says hg19, and the sequence still has the `<>` cut-site marker from CHOPCHOP. For the real run we removed the `<>` and switched to hg38, which the results page confirms. If you use this image in the paper, retake it with hg38 selected and no `<>`.

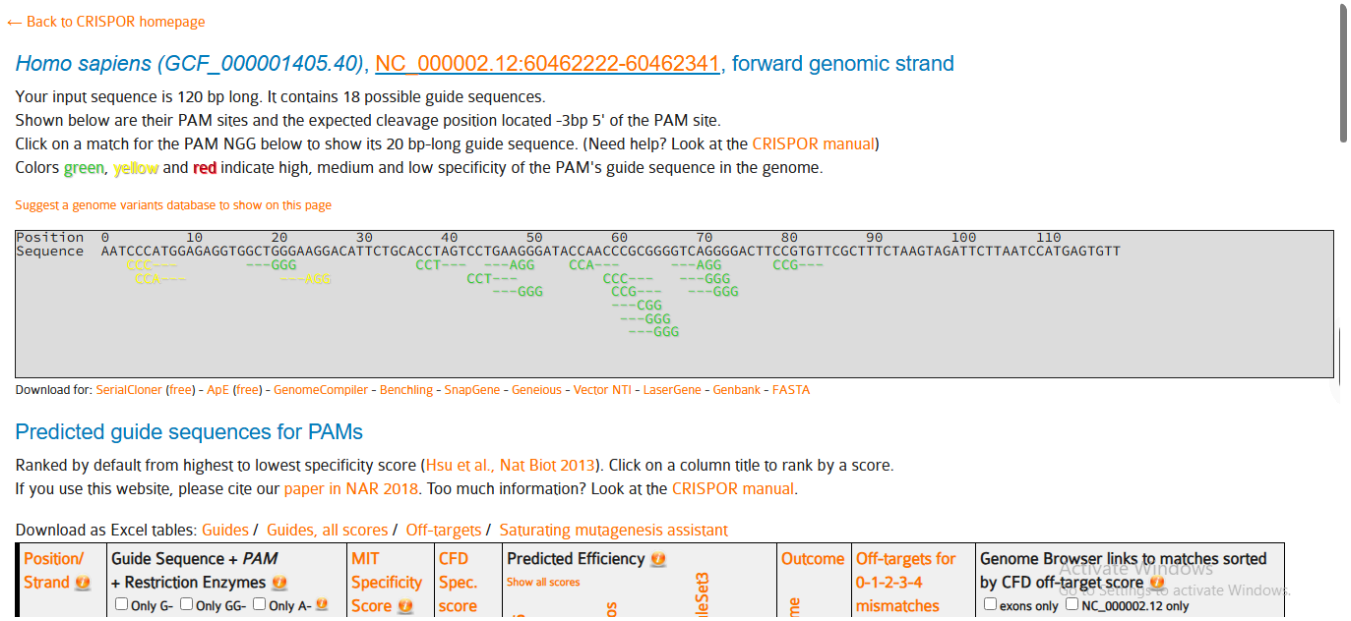
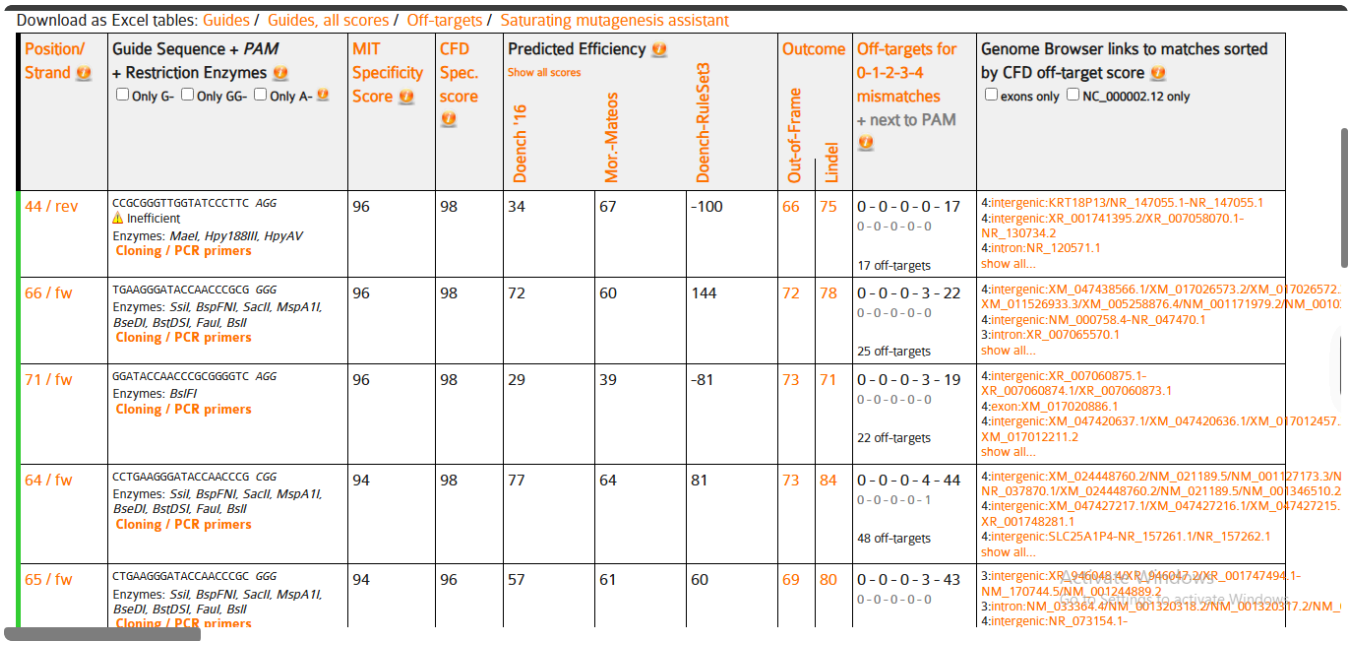


Figure 4B. CRISPOR's map of the 120 bp input.

What you are looking at: CRISPOR found 18 possible guides in our sequence. Each coloured label marks a PAM and the guide next to it. Green means high specificity, yellow medium, and red low. The page header shows the region NC_000002.12:60462222-60462341, which is hg38, and our guide starts at 60462265, exactly where CHOPCHOP put it. So both tools are looking at the same spot.



Position/ Strand	Guide Sequence + PAM + Restriction Enzymes	MIT Specificity Score	CFD Spec. score	Predicted Efficiency (Doench '16)	Predicted Efficiency (Mor.-Mateos)	Predicted Efficiency (Doench-RuleSet3)	Outcome (Out-of-Frame)	Outcome (Lindel)	Off-targets for 0-1-2-3-4 mismatches + next to PAM	Genome Browser links to matches sorted by CFD off-target score
44 / rev	CCGCGGGTTGGTATCCCTTC *AGG* ⚠ Inefficient Enzymes: *MaeI, Hpy188III, HpyAV* Cloning / PCR primers	96	98	34	67	-100	66	75	0 - 0 - 0 - 0 - 17 0 - 0 - 0 - 0 - 0 17 off-targets	4:intergenic:KRT18P13/NR_147055.1-NR_147055.1 4:intergenic:XR_001741395.2/XR_007058070.1-NR_130734.2 4:intron:NR_120571.1 show all...
66 / fw	TGAAGGGATACCAACCCGCG *GGG* Enzymes: *SsiI, BspFNI, SacII, MspA1I, BseDI, BstDSI, FauI, BslI* Cloning / PCR primers	96	98	72	60	144	72	78	0 - 0 - 0 - 3 - 22 0 - 0 - 0 - 0 - 0 25 off-targets	4:intergenic:XM_047438566.1/XM_017026573.2/XM_017026572.2/XM_011526933.3/XM_005258876.4/NM_001171979.2/NM_0010 4:intergenic:NM_000758.4-NR_047470.1 3:intron:XR_007065570.1 show all...
71 / fw	GGATACCAACCCGCGGGGTC *AGG* Enzymes: *BslFI* Cloning / PCR primers	96	98	29	39	-81	73	71	0 - 0 - 0 - 3 - 19 0 - 0 - 0 - 0 - 0 22 off-targets	4:intergenic:XR_007060875.1-XR_007060874.1/XR_007060873.1 4:exon:XM_017020886.1 4:intergenic:XM_047420637.1/XM_047420636.1/XM_017012457.2/XM_017012211.2 show all...
64 / fw	CCTGAAGGGATACCAACCCG *CGG* Enzymes: *SsiI, BspFNI, SacII, MspA1I, BseDI, BstDSI, FauI, BslI* Cloning / PCR primers	94	98	77	64	81	73	84	0 - 0 - 0 - 4 - 44 0 - 0 - 0 - 0 - 1 48 off-targets	4:intergenic:XM_024448760.2/NM_021189.5/NM_001127173.3/NR_037870.1/XM_024448760.2/NM_021189.5/NM_001346510.2 4:intergenic:XM_047427217.1/XM_047427216.1/XM_047427215.1/XR_001748281.1 4:intergenic:SLC25A1P4-NR_157261.1/NR_157262.1 show all...
65 / fw	CTGAAGGGATACCAACCCGC *GGG* Enzymes: *SsiI, BspFNI, SacII, MspA1I, BseDI, BstDSI, FauI, BslI* Cloning / PCR primers	94	96	57	61	60	69	80	0 - 0 - 0 - 3 - 43 0 - 0 - 0 - 0 - 0	3:intergenic:XR_946048.1/XR_946047.2/XR_001747494.1-NM_170744.5/NM_001244889.2 3:intron:NM_033364.4/NM_001320318.2/NM_001320317.2/NM_ 4:intergenic:NR_073154.1-

Figure 4C. CRISPOR scores for the guides. Our guide is the fourth row (64 / fw).

What you are looking at: Our guide `CCTGAAGGGATACCAACCCG` scores MIT 94 and CFD 98 for specificity, and 77 for efficiency (Doench '16). It has no off-targets with 0 to 2 mismatches, then 4 sites with 3 mismatches and 44 with 4 mismatches. The neighbouring guide in the second row is slightly more specific (MIT 96) but less efficient (72), so our guide trades a little specificity for more predicted efficiency.

For our guide, CRISPOR reported:

- MIT specificity: 94
- CFD specificity: 98
- Efficiency, Doench '16: 77 (Moreno-Mateos 64, Rule Set 3: 81)
- Off-targets at 0, 0, 0 mismatches for the first three categories, then 4 sites with 3 mismatches and 44 with 4 mismatches (48 in total)

5. Putting the two tools side by side

Check	CHOPCHOP	CRISPOR
Efficiency	76.99	Doench '16 = 77
Off-targets, 0-2 mismatches	none	none
Off-targets, 3 mismatches	1	4
Predicted frameshift	83.91%	Lindel 84

The two tools **agree on efficiency and on having no close off-targets**. They disagree a little on the number of 3-mismatch sites (1 versus 4), because they use different search settings. We report both numbers honestly rather than picking the smaller one.

One more honest point: this guide is not the top one on every CRISPOR score. A neighbouring guide (`TGAAGGGATACCAACCCGCG`) has a slightly better specificity score (MIT 96 against 94) and fewer total off-targets, but our guide has the higher predicted efficiency (77 against 72). In short, it trades a little specificity for more efficiency.

6. What this proves

We went from a gene name to a ready-to-use guide, primers and a predicted repair profile in a few clicks. We built **no protein at all**. That supports the central claim of the review: **CRISPR-Cas9 targeting is simple, because it only needs a ~20 letter spacer next to an**

NGG PAM.

7. Limits we must be upfront about

- Every number here is a **computer prediction**, not an experiment. Real editing depends on the cells, the delivery method and more.
- 83.91% means "of the predicted repair outcomes", **not** "83.91% of cells will be knocked out".
- The tools disagree on 3-mismatch off-target counts, so off-target risk should be tested in the lab (for example by sequencing).
- This is a coding-exon knock-out example, not the Casgevy enhancer approach.

8. Sources to cite (please verify before using)

These were given from memory, so check the journal, year and details against the original papers.

- CHOPCHOP: Labun et al., 2019, *Nucleic Acids Research*
- CRISPOR: Concordet & Haeussler, 2018, *Nucleic Acids Research*
- Repair-outcome prediction: Shen et al., 2018 (as named on the CHOPCHOP page)

9. Checklist before the section goes into the paper

- Crop out the "Activate Windows" watermark.
- Say "predicted" every time an efficiency or frameshift number is quoted.
- Add the access date (24 Oct 2026) and the genome version (hg38) to each figure caption.
- Keep the note that this is an illustrative design, not the therapeutic guide.